

Optimization for Machine Learning

机器学习中的优化方法

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Outline

- 1 Stochastic optimization
- 2 Stochastic gradient descent
- 3 Stochastic variance reduced gradient

Empirical risk minimization

Let $\{\mathbf{a}_i, b_i\}_{i=1}^n$ be n random samples. In machine learning, we usually learn model parameters $\mathbf{x}$ by optimizing

$$\min_{\mathbf{x} \in \mathbb{R}^d} F(\mathbf{x}) \triangleq \frac{1}{n} \sum_{i=1}^n f(\mathbf{x}; \{\mathbf{a}_i, b_i\}).$$

- hinge loss (support vector machine):

$$f(\mathbf{x}; \{\mathbf{a}_i, b_i\}) = \max\{1 - b_i \mathbf{a}_i^\top \mathbf{x}, 0\}$$

- logistic loss (logistic regression):

$$f(\mathbf{x}; \{\mathbf{a}_i, b_i\}) = \log(1 + \exp(-b_i \mathbf{a}_i^\top \mathbf{x}))$$

- neural network

Stochastic optimization

More generally, we consider the stochastic optimization problem

$$\min_{\mathbf{x} \in \mathbb{R}^d} F(\mathbf{x}) \triangleq \underbrace{\mathbb{E}_{\xi}[f(\mathbf{x}; \xi)]}_{\text{expectation setting}},$$

where the random variable $\xi \sim \mathcal{D}$.

- ξ is the randomness in problem.
- In this lecture, we suppose $F(\mathbf{x})$ is differentiable and convex.

Finite-sum setting

The finite-sum setting is a special case of the expectation setting:

$$F(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n f_i(\mathbf{x}).$$

If one draws index i from $\{1, 2, \dots, n\}$ uniformly at random, then

$$F(\mathbf{x}) = \mathbb{E}_i[f_i(\mathbf{x})].$$

A natural solution

Under “mild” assumptions, we have

$$\begin{aligned}\mathbf{x}_{t+1} &= \mathbf{x}_t - \eta_t \nabla F(\mathbf{x}_t) \\ &= \mathbf{x}_t - \eta_t \nabla \mathbb{E}[f(\mathbf{x}_t, \xi)] \\ &= \mathbf{x}_t - \eta_t \mathbb{E}[\nabla_{\mathbf{x}} f(\mathbf{x}_t, \xi)]\end{aligned}$$

issues:

- For the expectation setting, distribution of ξ may be unknown.
- For the finite-sum setting, computing full gradient is very expensive when n is very large.

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Stochastic gradient descent (SGD)

Stochastic gradient descent:

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \eta_t g(\mathbf{x}_t, \xi),$$

where $g(\mathbf{x}_t, \xi)$ is an **unbiased** estimator of $\nabla F(\mathbf{x}_t)$, i.e.,

$$\mathbb{E}[g(\mathbf{x}_t, \xi)] = \nabla F(\mathbf{x}_t).$$

For the finite-sum setting, we can choose index i_t from $\{1, 2, \dots, n\}$ uniformly at random. Then $\nabla f_{i_t}(\mathbf{x}_t)$ is an **unbiased** estimator of $\nabla F(\mathbf{x}_t)$:

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \eta_t \nabla f_{i_t}(\mathbf{x}_t)$$

Strongly convex and smooth problems

$$\min_{\mathbf{x} \in \mathbb{R}^d} F(\mathbf{x}) \triangleq \mathbb{E}_{\xi}[f(\mathbf{x}; \xi)]$$

Assumptions:

- $F(\mathbf{x})$ is L -smooth and μ -strongly convex (we do not require assumptions on $f(\mathbf{x}; \xi)$);
- Given $\xi_0, \dots, \xi_{t-1}$, $g(\mathbf{x}_t, \xi_t)$ is an unbiased estimator of $\nabla F(\mathbf{x}_t)$, i.e.,

$$\mathbb{E}[g(\mathbf{x}_t, \xi_t) | \xi_0, \dots, \xi_{t-1}] = \nabla F(\mathbf{x}_t);$$

- For all $\mathbf{x}$, we have $\underbrace{\mathbb{E}[\|g(\mathbf{x}, \xi)\|_2^2]}_{\text{bounded variance}} \leq \sigma^2$.

Convergence with fixed stepsizes

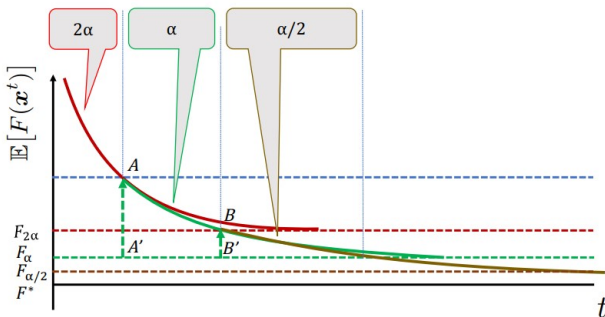
Under the assumptions in page 7, if $\eta_t = \eta \leq 1/(2L)$, then SGD achieves

$$\mathbb{E} \left[\|\mathbf{x}_t - \mathbf{x}^*\|_2^2 \right] \leq (1 - 2\eta\mu)^t \|\mathbf{x}_0 - \mathbf{x}^*\|_2^2 + \frac{\eta\sigma^2}{2\mu};$$

- fast (linear) convergence at the very beginning
- converges to some neighborhood of $\mathbf{x}^*$
- smaller stepsize η yield better converging points

One practical strategy

Run SGD with fixed stepsizes; whenever progress stalls, half the stepsize and continue SGD.



Convergence with diminishing stepsizes

Under the assumptions in page 7, if $\eta_t = \frac{\theta}{t+1}$ for some $\theta > \frac{1}{2\mu}$, then SGD achieves

$$\mathbb{E} \left[\|\mathbf{x}_t - \mathbf{x}^*\|_2^2 \right] \leq \frac{\alpha_\theta}{t+1}$$

where $\alpha_\theta = \max \left\{ \|\mathbf{x}_0 - \mathbf{x}^*\|_2^2, \frac{2\theta^2\sigma^2}{2\mu\theta-1} \right\}$.

Convex and smooth problems

$$\min_{\mathbf{x} \in \mathbb{R}^d} F(\mathbf{x}) \triangleq \mathbb{E}_{\xi}[f(\mathbf{x}; \xi)]$$

Assumptions:

- F is L -smooth and convex;
- Given $\xi_0, \dots, \xi_{t-1}$, $g(\mathbf{x}_t, \xi_t)$ is an unbiased estimator of $\nabla F(\mathbf{x}_t)$;
- For all $\mathbf{x}$, we have $\underbrace{\mathbb{E}[\|g(\mathbf{x}, \xi)\|_2^2]}_{\text{bounded variance}} \leq \sigma^2$.

Convex and smooth problems

Suppose we return a weighted average

$$\tilde{\mathbf{x}}_t = \sum_{k=0}^t \frac{\eta_k}{\sum_{j=0}^t \eta_j} \mathbf{x}_k$$

If F is convex, we have

$$\mathbb{E}[F(\tilde{\mathbf{x}}_t) - F(\mathbf{x}^*)] \leq \frac{\|\mathbf{x}_0 - \mathbf{x}^*\|_2^2 + \sum_{k=0}^t \sigma^2 \eta_k^2}{2 \sum_{k=0}^t \eta_k}.$$

If we choose $\eta_t = \Theta(1/\sqrt{t})$, we can get

$$\mathbb{E}[F(\tilde{\mathbf{x}}_t) - F(\mathbf{x}^*)] \leq O\left(\frac{\|\mathbf{x}_0 - \mathbf{x}^*\|_2^2 + \sigma^2 \log t}{\sqrt{t}}\right).$$

Convex and smooth problems

If we return

$$\tilde{\mathbf{x}}_t = \sum_{k=\lceil \frac{t}{2} \rceil}^t \frac{\eta_k \mathbf{x}_k}{\sum_{j=\lceil \frac{t}{2} \rceil}^t \eta_j}$$

Then we have

$$\mathbb{E}[F(\tilde{\mathbf{x}}_t) - F(\mathbf{x}^*)] \leq \frac{\|\mathbf{x}_0 - \mathbf{x}^*\|_2^2 + \sum_{k=\lceil \frac{t}{2} \rceil}^t \sigma^2 \eta_k^2}{2 \sum_{k=\lceil \frac{t}{2} \rceil}^t \eta_k} = O\left(\frac{\|\mathbf{x}_0 - \mathbf{x}^*\|_2^2 + \sigma^2}{\sqrt{t}}\right).$$

Comparisons with batch GD

$$\min_{\mathbf{x} \in \mathbb{R}^d} F(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n f_i(\mathbf{x}).$$

	iteration complexity	per-iteration	total
batch GD	$\kappa \log(1/\epsilon)$	n	$n\kappa \log(1/\epsilon)$
SGD	$1/\epsilon$	1	$1/\epsilon$

Table: Convergence rate for the strongly convex case

	iteration complexity	per-iteration	total
batch GD	$1/\epsilon$	n	n/ϵ
SGD	$1/\epsilon^2$	1	$1/\epsilon^2$

Table: Convergence rate for the convex case

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Stochastic variance reduced gradient (SVRG)

NOTE: For some $\mathbf{v}_t$ with $\mathbb{E}[\mathbf{v}_t] = \mathbf{0}$, $\mathbf{g}_t = \nabla f_{i_t}(\mathbf{x}_t) - \mathbf{v}_t$ is still an unbiased estimator of $\nabla F(\mathbf{x}_t)$.

If we have access to a history point $\tilde{\mathbf{x}}$ and $\nabla F(\tilde{\mathbf{x}})$, how to build a unbiased gradient estimator with converges to $\mathbf{0}$?

$$\underbrace{\nabla f_i(\mathbf{x}_t) - \nabla f_i(\tilde{\mathbf{x}})}_{\rightarrow \mathbf{0} \text{ if } \mathbf{x}_t \approx \tilde{\mathbf{x}}} + \underbrace{\nabla F(\tilde{\mathbf{x}})}_{\rightarrow \mathbf{0} \text{ if } \tilde{\mathbf{x}} \approx \mathbf{x}^*}$$

where i is randomly sampled from $\{1, \dots, n\}$.

- an unbiased estimator of $\nabla F(\mathbf{x}_t)$
- converges to $\mathbf{0}$ if $\mathbf{x}_t \approx \tilde{\mathbf{x}} \approx \mathbf{x}^*$

Stochastic variance reduced gradient (SVRG)

- operate in epochs
- in the s -th epoch
 - **beginning:** take a snapshot $\tilde{\mathbf{x}}$ of the current iterate, and compute the **batch gradient**

$$\nabla F(\tilde{\mathbf{x}}) = \frac{1}{n} \sum_{i=1}^n \nabla f_i(\tilde{\mathbf{x}}).$$

- **inner loop:** use the snapshot point to help reduce variance

$$\mathbf{x}_{t+1} = \mathbf{x}_t - \eta_t (\nabla f_i(\mathbf{x}_t) - \nabla f_i(\tilde{\mathbf{x}}) + \nabla F(\tilde{\mathbf{x}})).$$

Stochastic variance reduced gradient (SVRG)

Algorithm 1 Stochastic Variance Reduced Gradient

```
1: Input:  $\mathbf{x}_0, \eta, m, S$ 
2:  $\tilde{\mathbf{x}}^{(0)} = \mathbf{x}_0$ 
3: for  $s = 0, \dots, S - 1$ 
4:    $\mathbf{x}_0 = \tilde{\mathbf{x}}^{(s)}$ 
5:   for  $t = 0, \dots, m - 1$ 
6:     draw  $i_t$  from  $\{1, \dots, n\}$  uniformly at random
7:      $\mathbf{x}_{t+1} = \mathbf{x}_t - \eta(\nabla f_{i_t}(\mathbf{x}_t) - \nabla f_{i_t}(\tilde{\mathbf{x}}^{(s)}) + \nabla F(\tilde{\mathbf{x}}^{(s)}))$ ,
8:   end for
9:   Option I:  $\tilde{\mathbf{x}}^{(s+1)} = \mathbf{x}_m$ 
10:  Option II:  $\tilde{\mathbf{x}}^{(s+1)} = \mathbf{x}_t$  for randomly chosen  $t \in \{0, \dots, m - 1\}$ 
11: end for
12: Output:  $\tilde{\mathbf{x}}^{(S)}$ 
```

Remark

- constant stepsize η
- each epoch contains $2m + n$ gradient computations
- the average per-iteration cost of SVRG is comparable to that of SGD if $m \gtrsim n$

Convergence analysis

Suppose $F(\mathbf{x})$ is L -smooth and μ -strongly convex. Let $\eta = \Theta(1/L)$ and $m = \Theta(\kappa)$ is sufficient large so that

$$\rho = \frac{1}{\mu\eta(1 - 2L\eta)m} + \frac{2L\eta}{1 - 2L\eta} < 1,$$

then SVRG holds that

$$\mathbb{E}[F(\tilde{\mathbf{x}}^{(s)}) - F(\mathbf{x}^*)] \leq \rho^s (F(\mathbf{x}_0) - F(\mathbf{x}^*)).$$

To achieve

$$\mathbb{E}[F(\tilde{\mathbf{x}}^{(s)}) - F(\mathbf{x}^*)] \leq \epsilon$$

we only require at most $\mathcal{O}((n + \kappa) \log(1/\epsilon))$ number of gradient computations.

Important Lemma:

$$\mathbb{E}_t \left[\left\| \mathbf{g}_t^{(s)} \right\|_2^2 \right] \leq 4L \left[F(\mathbf{x}_t^{(s)}) - F(\mathbf{x}^*) + F(\tilde{\mathbf{x}}^{(s)}) - F(\mathbf{x}^*) \right]$$

Summary

$$\min_{\mathbf{x} \in \mathbb{R}^d} F(\mathbf{x}) = \frac{1}{n} \sum_{i=1}^n f_i(\mathbf{x}).$$

	iteration complexity	per-iteration	total
batch GD	$\kappa \log(1/\epsilon)$	n	$n\kappa \log(1/\epsilon)$
SGD	$1/\epsilon$	1	$1/\epsilon$
SVRG	$\log(1/\epsilon)$	$n + \kappa$	$(n + \kappa) \log(1/\epsilon)$

Table: Convergence rate for the strongly convex case